Arts et Métiers Sciences et Technologies
CER Aix-en-Provence

Smart sensors, actuators and manufacturing technologies
Arnaud Polette
arnaud.polette@ensam.eu

Lab Work – Machine Learning on Machining Data

The final deliverable must include a report explaining your methodology, as well as the code corresponding to each part. Your explanations must be illustrated with plots.

Part 1 : The database.

The database was generated during an acquisition campaign involving the machining of 169 holes drilled with a 6.36 mm drill bit (i.e., 6360 microns). For each hole, 25 measurements were taken on the drilling process (9 diameters, 14 roughness values, and 2 position measurements) and 13 during the process (6 cutting force measurements, 6 cutting torque measurements, and one energy measurement).

Diameter measurements :	**Roughness measurements :**
`D_In` : Average entry-hole diameter	`Rz_In_max` : Maximum Rz roughness at entry
`D_In_max` : Maximum entry-hole diameter	`Rz_In` : Average Rz roughness at entry
`D_In_min` : Minimum entry-hole diameter	`Rz_Out_max` : Maximum Rz roughness at exit
`D_Out` : Average exit-hole diameter	`Rz_Cly_max` : Maximum Rz roughness on hole wall
`D_Out_max` : Maximum exit-hole diameter	`Rz_Cyl` : Average Rz roughness on hole wall
`D_Out_min` : Minimum exit-hole diameter	`Rt_In` : Average Rt roughness at entry
`D_Cyl` : Average diameter on hole wall	`Rt_Out` : Average Rt roughness at exit
`D_Cyl_max` : Maximum diameter on hole wall	`Rt_Cyl` : Average Rt roughness on hole wall
`D_Cyl_min` : Minimum diameter on hole wall	`Rq_In` : Average Rq roughness at entry
Position measurements :	`Rq_Out` : Average Rq roughness at exit
`P(x)` : Hole position in x	`Rq_Cyl` : Average Rq roughness on hole wall
`P(y)` : Hole position in y	`Ra_In` : Average Ra roughness at entry
Energy measurement :	`Ra_Out` : Average Ra roughness at exit
`Et_Consommee` : Total energy consumed	`Ra_Cyl` : Average Ra roughness on hole wall

Cutting force measurements :	**Cutting torque measurements :**
`Moy_F_T1&T2` : Average cutting force between T1 and T2	`Moy_M_T1&T2` : Average cutting torque between T1 and T2
`Moy_F_T2&T3` : Average cutting force between T2 and T3	`Moy_M_T2&T3` : Average cutting torque between T2 and T3
`Moy_F_T3&T4` : Average cutting force between T3 and T4	`Moy_M_T3&T4` : Average cutting torque between T3 and T4
`Max_F_T1&T2` : Maximum cutting force between T1 and T2	`Max_M_T1&T2` : Maximum cutting torque between T1 and T2
`Max_F_T2&T3` : Maximum cutting force between T2 and T3	`Max_M_T2&T3` : Maximum cutting torque between T2 and T3
`Max_F_T3&T4` : Maximum cutting force between T3 and T4	`Max_M_T3&T4` : Maximum cutting torque between T3 and T4

The data is available in the file `data_capteurs.py` on SAVOIR. To access it, simply import it using `import data_capteurs`, then access the data using the structure `data_capteurs.data`. Here are several examples of how to use this structure :

- `data_capteurs.data["D_In"][0]` returns the average entry diameter of the first hole (index 0).
- `data_capteurs.data["Et_Consommee"]` returns the list of all energy consumption values.
- `data_capteurs.data["Moy_F_T1&T2"][41]` returns the average cutting force between T1 and T2 for hole index 41.
- `[data_capteurs.data["P(x)"][0], data_capteurs.data["P(y)"][0]]` : position of the first hole
- `etc ...`

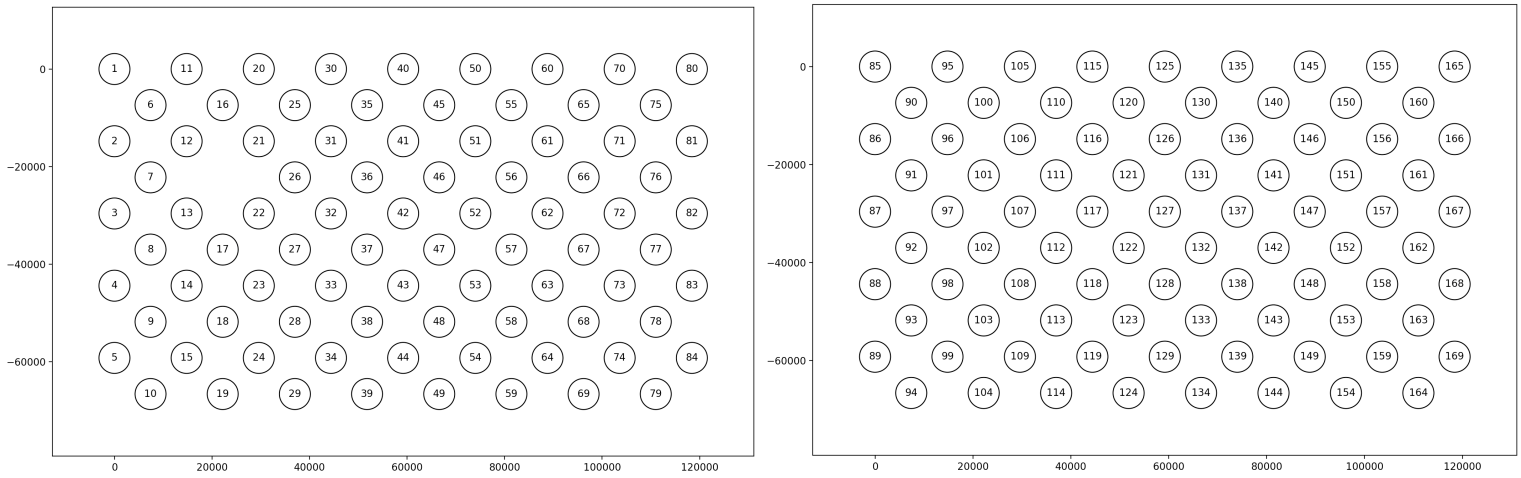


FIGURE 1 – Hole positions (`P(x)` and `P(y)`), on the two plates.

An example of data plotting is also provided on SAVOIR (`exemple.py`); it will give you a starting point to explore the database.

A first essential step is to become familiar with the database by manipulating it and understanding its contents.

Plot the different data series, using curves or any other type of graph you consider relevant to better understand this dataset.

Part 2 : Development of different problem statements.

This rich database offers many opportunities for data analysis and machine learning exploration. For each point, develop the problem using experiments, illustrating your explanations with graphs, calculations, learning results, or analyses from other methods. Because the amount of data is small, it will be necessary to use an appropriate validation method : cross-validation, which provides more relevant scores than standard validation when few data points are available (to discover/research independently).

1. During a drilling campaign with many holes, the drill bit gradually wears out, potentially causing increasing errors in the hole diameter. The first objective is to predict the evolution of the diameter error, knowing the number of holes already drilled with the bit. Try determining values within the experimental range, then values outside this range. What do you observe ?
2. Energy consumption and cutting force/torque information are strongly related. Can one be predicted from the other ? Considering that a greater force is required as the drill bit wears, and using the reduction of drilling diameter as a measure of wear, can we estimate this wear using energy consumption ?
3. We may assume that surface roughness increases with tool wear because the cut becomes less precise. Is it possible to predict roughness knowing the number of cuts already made ? By adding other information ?
4. Using N previous data points to predict the next one is a possible approach for various variables. This N value corresponds to the number of holes used to predict a subsequent value. The goal is to experiment with several N values over multiple characteristics, individually or on a set of variables.
5. Does the position of the hole on the plate influence the data ? How could we verify this ?
6. Bonus : propose an interesting study of your own on these data !